

Resume / Curriculum Vitae

Jason E. Orlosky

Primary: Augusta University, Department of Computer and Cyber Sciences
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Experience

Augusta University, School of Computer and Cyber Science: Associate Professor, October 2015 - Present

At Augusta University, I am conducting research on eye-tracked virtual reality as a step towards telehealth and the use of VR for improved education. Projects include the evaluation and diagnosis of patients with neurodegenerative diseases such as Parkinson's disease, simulations for post-stroke rehabilitation, and studying how simulation of visual deficits and experiential learning can affect student empathy towards patients.

Osaka University, Cybermedia Center: Specially Appointed Associate Professor, April 2017 - Present

At Osaka University, I am responsible for conducting cutting edge research in the fields of Augmented and Virtual Reality and Artificial Intelligence. My team consists of a small group of students ranging from bachelors to PhD level, who focus on head mounted display and interaction technologies. I am also actively promoting international collaboration between businesses and academic institutions by supporting joint projects and patents.

German Research Center for Artificial Intelligence (DFKI): Guest Researcher

October - November 2013 & January - March 2015

At DFKI, I conducted research on the combination of eye tracking and wearable augmented reality devices. In addition to developing a multi-focal plane head mounted display prototype, I explored the use of such technology for intelligent and automatic removal of distracting virtual content, as a memory augmentation device for people with dementia or memory loss, and for the development of on-demand vision augmentations.

McKesson Provider Technologies: Sales Engineer, January 2007 - March 2010

This role consisted of designing hospital wide IT solutions including networks, server configurations, and storage. I worked with hundreds of hospital systems in the U.S. and Canada, and customized and sold over \$8 million in hardware and services. Out of 25 peers in my group, I received the Engineer of The Year award for the 2008-2009 fiscal year. I also founded an internal educational program that teaches MBA curriculum to new hires, resulting in more effective corporate education and increased inter-department communication.

Georgia Tech Housing - ResNet: Residential Technical Advisor III, May 2003 - December 2006

As part of the residential network support team, I provided network and computer support to Georgia Tech students and staff, and tested and maintained on-campus local area networks. This included assisting with move-in network configurations and evening support to ensure students' computing needs were always met.

Georgia Tech, Department of Psychology: Sonification Research Assistant, January - December 2006

As a part time research assistant, I developed and tested sonification software that translates visual stimuli, such as fish in an aquarium, into sound, providing a new source of education for the visually impaired.

Education

Osaka University, Osaka, Japan, March 2013 - March 2016

Department of Information Systems Engineering - Ph.D. Information Science and Technology

Osaka University, Osaka, Japan, September 2011 - March 2013

Electrical Engineering Department - Exchange Program; Information Science Department - Research Student

University of Georgia, Athens, Georgia, January 2010 - May 2011

Franklin College of Arts and Sciences - Japanese Language and Literature (fluent in Japanese)

Georgia Institute of Technology, Atlanta, Georgia, August 2001 - December 2006

School of Electrical and Computer Engineering - B.S. Computer Engineering

Relevant Skills/Areas

Languages: Java, C++, C#, OpenCV, OpenGL (familiarity),

Tools: Eclipse, Visual Studio, Unity, GitHub, SourceTree, ARToolKit (familiarity), Matlab (familiarity)

Keywords: Augmented Reality, Virtual Reality, Intelligent User Interfaces, Vision Augmentation, Eye Tracking, Artificial Intelligence, Neurology

Field of Study

I am currently studying methods to improve usability and integration of wearable augmented reality (AR) and immersive Virtual Reality (VR) by incorporating a combination of artificial intelligence and user cognitive state. This includes prototyping input devices, developing algorithms to actively improve content viewability and management, and studying ways to augment human-machine interfaces and cognition. I have also recently started to explore neuroscience, eye tracking, and machine learning in more detail in an effort to help people with conditions such as Parkinson's and Alzheimer's. A detailed list of projects and papers can be found at www.JEOresearch.com/research.

Current and Planned Projects

Adaptive Augmented Reality – My primary PhD research was designed to improve the usability of and decrease invasiveness of virtual content. For example, when using a mobile display, overlaid text or objects can often be difficult to read, distract the user, and interfere with navigation tasks or interpersonal conversations. I have designed a number of systems to actively manage text placement and map content onto the real world to provide a more comfortable, efficient, and safe user experience.

Display Prototyping - To improve spatial interaction and field of view, I have prototyped two new head mounted displays as part of a collaborative team project with DFKI. The first is an optical see-through multi-focal plane display combined with an eye tracking interface. Through eye tracking and knowledge of the focal plane distances, we can calculate whether a user is looking at the environment or at a focal plane in the display. Any distracting text can then be quickly removed so that he or she has a clear view of the environment. The second prototype is a video see-through display that can manipulate a user's environmental view through fisheye and telescopic lenses, which can be engaged using eye gestures. Some of my current work involves improving these displays and testing them in the real world for use in applications like gaming and navigation.

Augmented Cognition – The goal of this project is to assist human cognition and processing through AR and VR. Though the research is still in its initial stages, I have proposed a framework to encode human memories into digital form for recall at a later time through eye-tracking and AR. Other work includes the use of eye tracking in VR to help diagnose or evaluate patients with neurodegenerative disorders through eye movements.

Past Research and Publications

Citations: 1799, h-index: 24, i10-index: 42

Peer Reviewed Journals

- 1) Hirobe, D., Shirai, S., Orlosky, J., Kobayashi, M., Uranishi, Y., and Takemura, H. (2026) TransPortal: A Portable Portal-Based Locomotion Technique. To appear in *IEEE Transactions on Visualization and Computer Graphics*.
- 2) Hirobe, D., Shirai, S., Orlosky, J., Alizadeh, M., Kobayashi, M., Uranishi, Y., Photchara, R., and Takemura, H. (2025) User-Centric Locomotion Techniques for Virtual Reality Games: A Survey of User Needs and Issues. *IEEE Transactions on Games*. 14 pages.
- 3) Li, Y. J., Orlosky, J., Jirau-Rosaly, W., Brown, S., & Rockich-Winston, N. (2024). Immersing healthcare students in a virtual reality Parkinson's disease experience. *Intl. Journal of Medical Education*, 15, 34-36.
- 4) Zhao, G., Orlosky, J., Gabbard, J., Kiyokawa, K. HazARdSnap: Gazed-based Augmentation Delivery for Safe Information Access while Cycling. (2023). *IEEE Transactions on Visualization and Computer Graphics*, 30(9), 6378-6389.
- 5) Sakamoto, K., Shirai, S., Takemura, N., Orlosky, J., Nagataki, H., Ueda, M., Uranishi, Y., and Takemura, H. (2023). Subjective Difficulty Estimation of Educational Comics Using Gaze Features. *The institute of Electronics, Information, and Communication Engineers (IEICE) Transactions on Information and Systems*, 106(5), 1038-1048.

- 6) Zhao, G., Orlosky, J., Feiner, S., Ratsamee, P., & Uranishi, Y. (2022). Mitigation of VR Sickness during Locomotion with a Motion-Based Dynamic Vision Modulator. *IEEE Transactions on Visualization and Computer Graphics*.
- 7) Orlosky, J., Sra, M., Bektaş, K., Peng, H., Kim, J., Kos' myna, N., Hollerer, T., Steed, A., Kiyokawa, K. and Akşit, K., (2021). Telelife: the future of remote living. *Frontiers in virtual reality*, 2, 763340.
- 8) Medeiros, A. C., Ratsamee, P., Orlosky, J., Uranishi, Y., Higashida, M., & Takemura, H. (2021). 3D pointing gestures as target selection tools: guiding monocular UAVs during window selection in an outdoor environment. *ROBOMECH Journal*, 8(1), 1-19.
- 9) Liu, C., Plopski, A., & Orlosky, J. (2020). OrthoGaze: Gaze-based Three-dimensional Object Manipulation using Orthogonal Planes. *Computers & Graphics*.
- 10) Orlosky, J., Theofilis, K., Kiyokawa, K., & Nagai, Y. (2020). Effects of Throughput Delay on Perception of Robot Teleoperation and Head Control Precision in Remote Monitoring Tasks. *PRESENCE: Virtual and Augmented Reality*, 27(2), 226-241.
- 11) Techasartikul, N., Ratsamee, P., Orlosky, J., Mashita, T., Uranishi, Y., Kiyokawa, K., & Takemura, H. (2020). Guidance and Visualization of Optimized Packing Solutions. *Journal of Information Processing*, 28, 193-202.
- 12) Techasartikul, N., Ratsamee, P., Orlosky, J., Mashita, T., Uranishi, Y., Kiyokawa, K., & Takemura, H. (2020). The Effect of the Presence of an Embodied Agent in an AR Guiding System. *Transactions of the Virtual Reality Society of Japan*, 25(1), 68-77.
- 13) Thomason, J., Ratsamee, P., Orlosky, J., Kiyokawa, K., Mashita, T., Uranishi, Y., & Takemura, H. (2019). A Comparison of Adaptive View Techniques for Exploratory 3D Drone Teleoperation. *ACM Transactions on Interactive Intelligent Systems (TiiS)*, 9(2-3), 17.
- 14) Ranchet, M., Orlosky, J., Morgan, J., Qadir, S., Akinwuntan, A. E., & Devos, H. (2017). Pupillary Response to Cognitive Workload during Saccadic Tasks in Parkinson's Disease. *Behavioural Brain Research*.
- 15) Orlosky, J., Itoh, Y., Ranchet, M., Kiyokawa, K., Morgan, J., Devos, H. (2017). Emulation of Physician Tasks in Eye-tracked Virtual Reality for Remote Diagnosis of Neurodegenerative Disease. *IEEE Transactions on Visualization and Computer Graphics (Proceedings of IEEEVR)*, 23(4), (pp. 1302-1311).
- 16) Orlosky, J., Toyama, T., Kiyokawa, K., Sonntag, D. (2015). Modular: Eye-controlled Vision Augmentations for Head Mounted Displays. In *Visualization and Computer Graphics, IEEE Transactions on (Proc. ISMAR) Vol. 21 (11)*. (pp. 1259-1268.) IEEE. **[Best Paper Nominee]**
- 17) Orlosky, J., Toyama, T., Sonntag, D., & Kiyokawa, K. (2015). The Role of Focus in Advanced Visual Interfaces. *KI-Künstliche Intelligenz*, (pp. 1-10).
- 18) Orlosky, J., Kiyokawa, K., Takemura, H. Managing Mobile Text in Head Mounted Displays: Studies on Visual Preference and Text Placement. In the *Mobile Computing and Communications Review*, Vol. 18 No. 2, 2014 (pp. 20-31).
- 19) Orlosky, J., Shigeno, T. Kiyokawa, K. Takemura, H. Text Input Evaluation with a Torso-mounted QWERTY Keyboard in Wearable Computing. *T. Virtual Reality Society of Japan. Vol.19, No.2*, 2014 (pp. 117-120).
- 20) Kishishita, N., Orlosky, J., Kiyokawa, K., Mashita, T., Takemura, H., (2014). "Investigation on the Peripheral Visual Field for Information Display with Wide-view See-through HMDs," *Transaction of the Virtual Reality Society of Japan. Vol.19, No.2*, 2014. (pp.121-130).

Invited Journal Papers

- 1) Orlosky, J. (2019) Current directions of HMD Research. *The Journal of the Virtual Reality Society of Japan*, Vol. 24 (3).
- 2) Orlosky, J., Kiyokawa, K., Takemura, H. (2017). Virtual and Augmented Reality on the 5G Highway. *Journal of Information Processing*, (25), pp. 133-141.

Peer Reviewed Conferences

- 1) Taheritajar, A., Zhao, J., & Orlosky, J. (2026, January). Augmented Reality Visual Retrieval for Object Detection and Corpus-Guided Content Generation. In *2026 IEEE International Conference on Artificial Intelligence and eXtended and Virtual Reality (AIxVR)* (pp. 48-56). IEEE.
- 2) Taheritajar, A., Benson, J., Gibson, A., Wilburn, B., Zhao, J. and Orlosky, J. (2025, October). Scalable Object Detection in Mixed Reality using Incremental Re-training and One-shot 3D Annotation. To appear in the Proceedings of the 2025 International Symposium on Mixed and Augmented Reality. IEEE.
- 3) Zhao, G., Orlosky, J., Kiyokawa, K., & Uranishi, Y. (2024, October). GlanXR: A Hands-Free Fast Switching System for Virtual Screens. In *2024 IEEE International Symposium on Mixed and Augmented Reality (ISMAR)* (pp. 111-119). IEEE. **[Best Paper Nominee]**
- 4) Zhao, G., Hu, X., Orlosky, J., & Kiyokawa, K. (2024, June). ReAR Indicators: Peripheral Cycling Indicators for Rear-Approaching Hazards. In *Proceedings of the 2024 International Conference on Advanced Visual Interfaces* (pp. 1-9).
- 5) Orlosky, J., Liu, C., Sakamoto, K., Sidenmark, L., & Mansour, A. (2024, March). EyeShadows: Peripheral Virtual Copies for Rapid Gaze Selection and Interaction. In *2024 IEEE Conference Virtual Reality and 3D User Interfaces (VR)* (pp. 681-689). IEEE. **[Honorable Mention for Best Paper Presentation]**
- 6) Hirobe, D., Uranishi, Y., Orlosky, J., Shirai, S., Ratsamee, P., & Takemura, H. (2022). Characteristics of Background Color Shifts Caused by Optical See-Through Head-Mounted Displays. In *Proceedings of the 32nd International Conference on Artificial Reality and Telexistence*. (pp. 139-147).
- 7) Huynh, B., Wysopal, A., Ross, V., Orlosky, J., & Höllerer, T. (2022, October). Layerable Apps: Comparing Concurrent and Exclusive Display of Augmented Reality Applications. In *2022 IEEE International Symposium on Mixed and Augmented Reality (ISMAR)* (pp. 857-863). IEEE.
- 8) Lim, D., Shirai, S., Orlosky, J., Ratsamee, P., Uranishi, Y., & Takemura, H. (2022, December). Evaluation of User Interfaces for Three-Dimensional Locomotion in Virtual Reality. In *Proceedings of the 2022 ACM Symposium on Spatial User Interaction* (pp. 1-9).
- 9) Orlosky, J., & Grabowski, T. (2021, June). Genetic crossover in the evolution of time-dependent neural networks. In *Proceedings of the Genetic and Evolutionary Computation Conference* (pp. 885-891).
- 10) Liu, C., Orlosky, J., & Plopski, A. (2020, October). Eye Gaze-based Object Rotation for Head-mounted Displays. In *Symposium on Spatial User Interaction* (pp. 1-9).
- 11) Orlosky, J., Huynh, B., & Hollerer, T. (2019, December). Using Eye Tracked Virtual Reality to Classify Understanding of Vocabulary in Recall Tasks. In *2019 IEEE International Conference on Artificial Intelligence and Virtual Reality (AIVR)* (pp. 66-667). IEEE.
- 12) Orlosky, J., Ezenwoye, O., Yates, H., & Besenyi, G. (2019, April). A Look at the Security and Privacy of Fitbit as a Health Activity Tracker. In *Proc. of the 2019 ACM Southeast Conference* (pp. 241-244). ACM.
- 13) Techasartikul, N., Ratsamee, P., Orlosky, J., Mashita, T., Uranishi, Y., Kiyokawa, K., & Takemura, H. (2019). Evaluation of Embodied Agent Positioning and Moving Interfaces for an AR Virtual Guide. *ICAT-EGVE 2019 - International Conference on Artificial Reality and Telexistence and Eurographics Symposium on Virtual Environments*.
- 14) Liu, C., Plopski, A., Kiyokawa, T., Ratsamee, P., & Orlosky, J. (2018). IntelliPupil: Pupillometric Light Modulation for Optical See-through Head Mounted Displays. To Appear in *In Mixed and Augmented Reality (ISMAR), 2018 IEEE International Symposium on*. IEEE.
- 15) Orlosky, J., Kim, P., Kiyokawa, K., Mashita, T., Ratsamee, P., Uranishi, Y., & Takemura, H. (2017, October). VisMerge: Light Adaptive Vision Augmentation via Spectral and Temporal Fusion of Non-visible Light. In *Mixed and Augmented Reality (ISMAR), 2017 IEEE International Symposium on* (pp. 22-31). IEEE.

- 16) Yano, Y., Orlosky, J., Kiyokawa, K., Takemura, H. (2016). Dynamic view expansion for improving visual search in video see-through AR. In *Proceedings of the 26th International Conference on Artificial Reality and Telexistence and the 21st Eurographics Symposium on Virtual Environments* (pp. 57-60).
- 17) Theofilis, K., Orlosky, J., Nagai, Y., and Kiyokawa, K., (2016) Panoramic view reconstruction for stereoscopic teleoperation of a humanoid robot. In *Humanoid Robots (Humanoids), 2016 IEEE-RAS 16th International Conference on* (pp. 242-248). IEEE.
- 18) Plopski, A., Orlosky, J., Itoh, Y., Nitschke, C., Kiyokawa, K., Klinker, G. (2016). Automated Spatial Calibration of HMD Systems with Unconstrained Eye-cameras. In *Mixed and Augmented Reality (ISMAR), 2016 IEEE International Symposium on* (pp. 94-99).
- 19) Itoh, Y., Orlosky, J., Kiyokawa, K., & Klinker, G. (2016). Laplacian Vision: Augmenting Motion Prediction via Optical See-Through Head-Mounted Displays. In *Proceedings of the 7th Augmented Human International Conference 2016* (p. 16). ACM. [\[Best Paper\]](#)
- 20) Orlosky, J., Kiyokawa, K., Toyama, T., & Sonntag, D. (2015). Halo Content: Context-aware Viewspace Management for Non-invasive Augmented Reality. In *the 20th International Conference on Intelligent User Interfaces* (pp. 369-373). ACM.
- 21) Toyama, T., Orlosky, J., Sonntag, D., & Kiyokawa, K. (2015). Attention Engagement and Cognitive State Analysis for Augmented Reality Text Display Functions. In *the 20th International Conference on Intelligent User Interfaces* (pp. 322-332). ACM.
- 22) Kishishita, N., Kiyokawa, K., Orlosky, J., Mashita, T., Takemura, H., & Kruijff, E. (2014). Analysing the effects of a wide field of view augmented reality display on search performance in divided attention tasks. In *Mixed and Augmented Reality (ISMAR), 2014 IEEE International Symposium on* (pp. 177-186). IEEE.
- 23) Orlosky, J., Wu, Q., Kiyokawa, K., Takemura, H., & Nitschke, C. (2014). Fisheye vision: peripheral spatial compression for improved field of view in head mounted displays. In *the 2nd ACM Symposium on Spatial User Interaction* (pp. 54-61). ACM.
- 24) Toyama, T., Orlosky, J., Sonntag, D., Kiyokawa, K. (2014). A natural interface for multi-focal plane head mounted displays using 3D gaze. In *the working conference on Advanced visual interfaces*, 2014 (pp. 25-32).
- 25) Orlosky, J., Kiyokawa, K., Takemura, H. (2013). Dynamic Text Management for See-through Wearable and Heads-up Display Systems. In *the 2013 International Conference on Intelligent User Interfaces*. (pp. 363-370). ACM. [\[Best Paper\]](#)

Peer Reviewed Posters, Demos, Consortia, and Workshops

- 1) Maruyama, Y., Orlosky, J., Lee, C., Salim, F., Starner, T., & Tag, B. (in press). Augmenting Mandarin Chinese pronunciation through tone visualisation in wearable glasses. To appear in Interspeech 2026, Show and Tell Track.
- 2) Elliott, R., Benson, J., Rad, P., & Orlosky, J. (2026, January). 3D View Alignment and Autonomous Navigation of Telepresence Robots During Multi-Participant Interaction in Augmented Reality. In *2026 IEEE International Conference on Artificial Intelligence and eXtended and Virtual Reality (AIxVR)* (pp. 248-252). IEEE.
- 3) Erb, L., & Orlosky, J. (2024, March). Tremor Stabilization for Sculpting Assistance in Virtual Reality. In *2024 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW)* (pp. 753-754). [\[Best Poster\]](#)
- 4) Koji, M., Uranishi, Y., Kiyokawa, K., Orlosky, J., Ratsamee, P., Kobayashi, M. (2023). Virtual Zoomorphic Accessories for Enhancing Perception of Vehicle Dynamics in Real-Time. *Proceedings of the 32nd International Conference on Artificial Reality and Telexistence*, pp. 41-42.
- 5) Zhao, G., Hu, X., Orlosky, J., Kiyokawa, K. (2023). ReAR – A Peripheral Indicator for Rear-Approaching Vehicles While Cycling. To appear in the 2023 International Symposium on Mixed and Augmented Reality Adjunct (ISMAR-Adjunct).
- 6) Mansour, A., & Orlosky, J. (2023, October). Approximated Match Swiping: Exploring more Ergonomic Gaze-based Text Input for XR. In *2023 IEEE International Symposium on Mixed and Augmented Reality*

Adjunct (ISMAR-Adjunct) (pp. 141-145). IEEE.

- 7) Bektaş, K., Kim, J., Peng, H., Kiyokawa, K., Steed, A., Höllerer, T., Kos'myna, N., Sra, M., Orlosky, J. and Akşit, K. (2022, May). Telelife: A vision of remote living in 2035. In *Extended Abstracts of the 2022 CHI Conference on Human Factors in Computing Systems, CHI* (Vol. 22).
- 8) Shaltout, N., Monteiro, D. V., Perusquía-Hernández, M., Orlosky, J., & Kiyokawa, K. (2022). How Anxiety State and Acceptance of an Embodied Agent Affect User Gaze Patterns. In *CEUR Workshop* (Vol. 3297).
- 9) Lim, D., Shirai, S., Orlosky, J., Ratsamee, P., Uranishi, Y., & Takemura, H. (2022, October). Exploring Three-Dimensional Locomotion Techniques in Virtual Reality. In *2022 International Symposium on Mixed and Augmented Reality Adjunct (ISMAR-Adjunct)* (pp. 501-502).
- 10) Hirobe, D., Uranishi, Y., Orlosky, J., Shirai, S., Ratsamee, P., & Takemura, H. (2022, October). Objective Measurements of Background Color Shifts Caused by Optical See-Through Head-Mounted Displays. In *IEEE International Symposium on Mixed and Augmented Reality Adjunct (ISMAR-Adjunct)* (pp. 389-390).
- 11) Takahashi, R., Shirai, S., Orlosky, J., Uranishi, Y., & Takemura, H. (2021, October). A Japanese Character Flick-Input Interface for Entering Text in VR. In *2021 IEEE International Symposium on Mixed and Augmented Reality Adjunct (ISMAR-Adjunct)* (pp. 251-253). IEEE.
- 12) Zhao, G., Orlosky, J., & Uranishi, Y. (2021, March). Evaluating Presence in VR with Self-Representing Auditory-Vibrotactile Input. In *2021 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW)* (pp. 577-578). IEEE.
- 13) Yoshimi, K., Ratsamee, P., & Orlosky, J. (2021). AREarthQuakeDrill: Toward Increased Awareness of Personnel during Earthquakes via AR Evacuation Drills. In *2021 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW)* (pp. 446-447). IEEE.
- 14) Wei, Y., Orlosky, J., & Mashita, T. (2021). Visualization and Manipulation of Air Conditioner Flow via Touch Screen. In *2021 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW)* (pp. 430-431). IEEE.
- 15) Huynh, B., Orlosky, J., & Höllerer, T. (2020). Designing a Multitasking Interface for Object-aware AR applications. In *2020 IEEE International Symposium on Mixed and Augmented Reality Adjunct (ISMAR-Adjunct)* (pp. 39-40). IEEE.
- 16) Sakamoto, K., Shirai, S., Orlosky, J., Nagataki, H., Takemura, N., Alizadeh, M., & Ueda, M. (2020). Exploring Pupillometry as a Method to Evaluate Reading Comprehension in VR-based Educational Comics. In *2020 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW)* (pp. 422-426). IEEE.
- 17) Alexander, D., Nguyen, T., Keller, P., Orlosky, J., Brown, S., Wood, E., Ezenwoye, O and Jirau-Rosaly, W. (2020). Design of Visual Deficit Simulation for Integration into a Geriatric Physical Diagnosis Course. To appear in the 2020 IEEE Conference on Virtual Reality and 3D User Interfaces. IEEE.
- 18) Goncalves, P., Machulla, T. and Orlosky, J., (2020). An Augmented Reality Assistant to Support Button Selection for Patients with Age-related Macular Degeneration. To appear in the 2020 IEEE Conference on Virtual Reality and 3D User Interfaces. IEEE.
- 19) Sakamoto, K., Shirai, S., Orlosky, J., Nagataki, H., Takemura, N., Alizadeh, M., Ueda, M. (2020). Exploring Pupillometry as a Method to Evaluate Reading Comprehension in VR-based Educational Comics. To appear in *KELVAR: 4th Annual Workshop on K-12+ Embodied Learning through Virtual and Augmented Reality2020*, IEEE Conference on Virtual Reality and 3D User Interfaces. IEEE.
- 20) Orlosky, J., Liu, C., Kalkofen, D., & Kiyokawa, K. (2019, October). Visualization-Guided Attention Direction in Dynamic Control Tasks. In *2019 IEEE International Symposium on Mixed and Augmented Reality Adjunct (ISMAR-Adjunct)* (pp. 372-373). IEEE.
- 21) Techarantikul, N., Mashita, T., Ratsamee, P., Uranishi, Y., Takemura, H., Orlosky, J., & Kiyokawa, K. (2019, March). Evaluation of Pointing Interfaces with an AR Agent for Multi-section Information Guidance.

In 2019 *IEEE Conference on Virtual Reality and 3D User Interfaces* (VR) (pp. 1185-1186). IEEE.

- 22) Zhang, H., & Orlosky, J. (2019, March). Thermal HDR: Applying High Dynamic Range Rendering for Fusion of Thermal Augmentations with Visible Light. In 2019 *IEEE Conference on Virtual Reality and 3D User Interfaces* (VR) (pp. 1263-1264). IEEE.
- 23) Hartney, J. H., Rosenthal, S. N., Kirkpatrick, A. M., Skinner, J. M., Hughes, J., & Orlosky, J. (2019, March). Revisiting Virtual Reality for Practical Use in Therapy: Patient Satisfaction in Outpatient Rehabilitation. In 2019 *IEEE Conference on Virtual Reality and 3D User Interfaces* (VR) (pp. 960-961). IEEE.
- 24) Huynh, B., Orlosky, J., & Höllerer, T. (2019, March). In-Situ Labeling for Augmented Reality Language Learning. In 2019 *IEEE Conference on Virtual Reality and 3D User Interfaces* (VR) (pp. 1606-1611). IEEE.
- 25) Huynh, B., Orlosky, J., & Höllerer, T. (2019, March). Semantic Labeling and Object Registration for Augmented Reality Language Learning. In 2019 *IEEE Conference on Virtual Reality and 3D User Interfaces* (VR) (pp. 986-987). IEEE.
- 26) Kim, P., Orlosky, J., Kiyokawa, K., Ratsamee, P., & Mashita, T. (2017, October). DotWarp: Dynamic Object Timewarping for Video See-Through Augmented Reality. In *Mixed and Augmented Reality (ISMAR-Adjunct), 2017 IEEE International Symposium on* (pp. 184-185). IEEE.
- 27) Mori, M., Orlosky, J., Kiyokawa, K., Takemura, H. (2016). A Transitional AR Furniture Arrangement System with Automatic View Recommendation. In *Mixed and Augmented Reality (ISMAR-Adjunct), 2016 IEEE International Symposium on* (pp. 158-159).
- 28) Orlosky, J., Weber, M., Gu, Y., Sonntag, D., & Sosnovsky, S. (2015). An Interactive Pedestrian Environment Simulator for Cognitive Monitoring and Evaluation. In *the 20th International Conference on Intelligent User Interfaces Companion* (pp. 57-60). ACM.
- 29) Orlosky, J., Depth based interaction and field of view manipulation for augmented reality. (2014). In *the adjunct publication of the 27th annual ACM symposium on User interface software and technology* (pp. 5-8).
- 30) Orlosky, J., Toyama, T., Sonntag, D., Sarkany, A., & Lorincz, A. (2014). On-body multi-input indoor localization for dynamic emergency scenarios: fusion of magnetic tracking and optical character recognition with mixed-reality display. In *Pervasive Computing and Communications Workshops (PERCOM Workshops), 2014 IEEE International Conference on* (pp. 320-325). IEEE.
- 31) Orlosky, J., Toyama, T., Sonntag, D., and Kiyokawa, K. (2014) "Using Eye-Gaze and Visualization to Augment Memory." In *Distributed, Ambient, and Pervasive Interactions*, pp. 282-291. Springer.
- 32) Orlosky, J., Kiyokawa, K., Takemura, H. Towards Intelligent View Management: A Study of Manual Text Placement Tendencies in Mobile Environments Using Video See-through Displays. In *Mixed and Augmented Reality (ISMAR), 2013 IEEE International Symposium on* (pp. 281-282). IEEE.
- 33) Orlosky, J., Kiyokawa, K., & Takemura, H. Management and Manipulation of Text in Dynamic Mixed Reality Workspaces, In *the 2013 International Symposium on Mixed and Augmented Reality, Doctoral Consortium*, (pp. 1-4) IEEE.
- 34) Kishishita, N., Orlosky, J., Mashita, T., Kiyokawa, K., Takemura, H. (2013, March). Investigation on the Peripheral Visual Field for Information Display with Real and Virtual Wide Field-of-view See-through HMDs. In *3D User Interfaces (3DUI), 2013 IEEE Symposium on* (pp. 143-144). IEEE.
- 35) Walker, B., Godfrey, M., Orlosky, J., Bruce, C., Sanford, J. (2006). Aquarium Sonification: Soundscapes for Accessible Dynamic Informal Learning Environments. *Proceedings of The 12th International Conference on Auditory Display*. (pp. 238-241).

Non-peer Reviewed Work

- 1) Kouame, G., Wood, E., & Orlosky, J. (2021). A Virtual Reality Library Space for Health Centered Education and Well-Being. *Journal of Hospital Librarianship*, 1-8.
- 2) Ratsamee, P., Uranishi, Y., Orlosky, J., Higashida, M., & Takemura, H. (2021). SmartVP: An autonomous

Viewpoint Selection for Watching a Boxing Game in Virtual Reality. システム制御情報学会研究発表講演会講演論文集, 65, 276-279.

- 3) Orlosky, J. (2017, June). Toward Parallel Consciousness: Classifying User State to Improve Augmentation Relevance. In *Ubiquitous Virtual Reality (ISUVR), 2017 International Symposium on* (pp. 34-37). IEEE.
- 4) Yates, H.A., Winslow, D.R., Orlosky, J., Ezenwoye, O. Besenyi, G.M. (in press). The future of active living big data: Validity and reliability of wearable activity trackers. Accepted for poster presentation at *the 14th Annual Active Living Research Conference*, February 11 – February 14, 201, Banff, Alberta, Canada.
- 5) Patel, S., Orlosky, J., Stewart, J., Coughlin, S.S., Besenyi, G.M. (in press). FitBit wearable devices: Accessing physical activity data for health promotion and intervention research. Accepted for oral presentation at the American Public Health Association 145th Annual Meeting and Exposition, Nov 4-8, 2017, Atlanta, GA.
- 6) Yates, H., Orlosky, J., Ezenwoye, O., Besenyi, G.M. (in press). Physical activity wearable technology: Assessing security and privacy risks for use in health research. Accepted for oral presentation at the American Public Health Association 145th Annual Meeting and Exposition, Nov 4-8, 2017, Atlanta, GA.
- 7) Orlosky, J., Adaptive Display for Augmented Reality. (2015, July). The 9th Young Researcher's Retreat, Nara, Japan. [\[Best Poster\]](#)
- 8) Sonntag, D., Orlosky, J., Weber, M., Gu, Y., Sosnovsky, S., Toyama, T., & Toosi, E. N. (2015, June). Cognitive Monitoring via Eye Tracking in Virtual Reality Pedestrian Environments. In *Proceedings of the 4th International Symposium on Pervasive Displays* (pp. 269-270). ACM.
- 9) Voros, G., Miksztaí Rethy, B., Vero, A., Orlosky, J., Toyoma, T., Sonntag, D., Lorincz. A., (2014). Mobile AAC Solutions using Gaze Tracking and Optical Character Recognition. In *the 16th Biennial Conference of the International Society for Augmentative and Alternative Communication (ISAAC)*.
- 10) Vero, A., B. Miksztaí Rethy, B., Pinter, B., Voros, G., Orlosky, J., Toyoma, T., Sonntag, D., Lorincz. A. (2014). Gaze Tracking and Language Model for Flexible Augmentative and Alternative Communication in Practical Scenarios. In *the 16th Biennial Conference of the International Society for Augmentative and Alternative Communication (ISAAC)*.
- 11) Orlosky, J., Kiyokawa, K., Takemura, H. Automated Text Management for Wearable and See-through Display Systems. (2013). *Workshop at The 6th Korea-Japan Workshop on Mixed Reality*, Okinawa, Japan.
- 12) Orlosky, J., Kiyokawa, K., Takemura, H. Scene Analysis for Improving Visibility in Wearable Displays. (2013). In *The 16th Meeting on Image Recognition and Understanding*, Tokyo, Japan.
- 13) Orlosky, J., Katzakis, N. Kiyokawa, K. Takemura, H. (2012). Torso Keyboard: A Wearable Text Entry Device That Can Be Used While Sitting, Standing or Walking. In *the 10th Asia Pacific Conference on Human Computer Interaction*. (pp. 781-782).

Doctoral Thesis

Adaptive Display of Virtual Content for Improving Usability and Safety in Mixed and Augmented Reality. Defended October 5th, 2015, final revision submitted January 5th, 2016.

Successful Grants, Sponsorship, and Licensing

August 2026 - July 2029 Co-I./Mentor National Science Foundation \$159,000 (100% Orlosky/Benson)
Graduate Research Fellowship Program: Awarded to PhD student Jeff Benson under my supervision.

2026 Augusta University Lic-2025-028: Commercial License for “Scalable object detection in mixed reality using incremental re-training and one-shot 3D annotation” and derivative work

2024 Intermural Grants Program P.I. Generation of Real-time AR Work Instructions using Object Recognition and Large Language Models - \$30,000 USD (100% Orlosky)

Spring 2026 Senior Design Program Sponsorship - \$10,000 USD

Fall 2025 Senior Design Program Sponsorship - \$10,000 USD
Spring 2025 Senior Design Program Sponsorship - \$25,000 USD
Fall 2024 Senior Design Program Sponsorship - \$20,000 USD
Summer 2024 Senior Design Program Sponsorship - \$5,000 USD
Spring 2024 Senior Design Program Sponsorship - \$35,000 USD
Fall 2023 Senior Design Program Sponsorship - \$15,000 USD
(Arranged for and administered sponsorship of Senior Capstone Projects through industry partner gifts.)

September 2022 - August 2025 Co-I. National Science Foundation \$499,000 USD (50% Orlosky)
Data-Driven Modeling and Control of Human-Cyber-Physical Systems with Extended-Reality Interfaces

June 2020 - June 2023 Co-I. Office of Naval Research \$52,118 USD (100% Orlosky)
Virtually Co-located Augmented Reality Spaces for Visualization, Training, and Navigation

April 2021 - March 2025 P.I. - Japan Society for the Promotion of Science ~\$156,340 USD (100% Orlosky)
Development of Cognitive Symbiosis in Virtual Agents to Improve Remote Classroom Learning Outcomes

November 2020 - March 2021 P.I. - UCL-Osaka University Strategic Partner Funds ~\$5,500 USD
Telelife Technology Seminars via Virtual Reality
July 2019 - June 2021 Co-I. - Japan Society for the Promotion of Science ~\$44,000 USD
Creation of an Adaptive Learning Environment based on Biological Information Analysis when Learning with Comic Books

April 2018 - March 2021 P.I. - Office of Naval Research \$372,980 USD
Vision Augmentation using Eye Tracked Augmented Reality

April 2015 - March 2017 P.I. - Japan Society for the Promotion of Science ~\$67,000 USD
Intelligent Text Overlay for Head Mounted Display Systems

Pending Grant Submissions

2026 NSF TTP-T: Context-aware Mixed Reality Workflows for Industrial Training, Maintenance and Process Control (P.I., \$994,512) (34% Orlosky)
2025 NSF CSSI: Collaborative Research: Frameworks: MR-Path: A Mixed-Reality and AI-Enabled Infrastructure for Advancing Annotating, Modeling, Sharing, and Educating in Digital Pathology (Co-I., \$265,825) (100% Orlosky)
2025 NSF VINES: TeamAR-X: Context-Aware, AI-Orchestrated Collaborative Augmented Reality over Next-Generation Networks (P.I. for Augusta Portion, \$239,729) (100% Orlosky)
2025 DoD Small Business Innovative Research Grant: Railsight AR (Co-I. \$76,652) (100% Orlosky)

Submitted (but not awarded) Grants and Sponsorship

2025 DoD Small Business Innovative Research Grant: ShipLens: Beyond Boundaries (\$71,999)
2025 NSF Core: Integration of Large Language Models, Object Detection, and Interactive Agents to Support Secondary Language Learning (\$566,444)
2024 DoD Small Business Innovative Research Grant: ATLAS: Augmented Training and Learning Assistance System (\$75k)
2024 NSF SBIR Phase I: Generation of Dynamic Content for Augmented Reality based Work Instructions for Manufacturing Processes using Conversational AI and Computer Vision (\$105k)
2024 Planning: CIVIC-FA Track B: Building a Digital On-Farm Experimentation Library for Urban Farmers. (\$3k)
2023 Public Safety and Community Violence Reduction Grant: Technology-facilitated Training and Research on De-escalation Strategies in an Adaptable Virtual Space (\$1.1M)
2023 Industry Sponsored Research - Diagnamed Telediagnosis System (\$30k, abandoned by sponsor)

2023 ONR Special Notice Whitepaper (\$900K)
 2023 Standardization of Calibration, Interaction, and Measurement Tools for Gaze-Enabled Augmented and Virtual Reality: ONR Whitepaper
 2022 ONR Young Investigator Award (\$500K)
 2022 Joint NIH submission with Dr. Gokila Dorai (\$500k)
 2021 Whitepaper submission to ONR Human Performance grant
 2021 Submission to Google Research Scholar Award (\$60k)
 2021 Joint NSF whitepaper submission with Dr. Ahmed Aleroud
 2021 NVIDIA Hardware program submission (\$10k)
 2021 Submission to ONR special notice grant (\$600k)

Teaching

2026 Summer CSCI 8940 Dissertation Research (4 students)
 2026 Spring CSCI 8940 Dissertation Research (4 students)
 2026 Spring CSCI 4712 Senior Capstone, primary instructor (28 students)
 2025 Fall CSCI 4712 Senior Capstone, primary instructor (23 students)
 2025 Fall CSCI 8940 Dissertation Research (4 students)
 2025 Summer CSCI 8940 Dissertation Research (4 students)
 2025 Spring CSCI 8940 Dissertation Research, primary instructor (4 students)
 2025 Spring CSCI 4712 Senior Capstone, primary instructor (24 students)
 2025 Spring CSCI 4990 Undergraduate Research, primary instructor (1 student)
 2024 Fall CSCI 4990 Undergraduate Research, primary instructor (4 students)
 2024 Fall CSCI 4712 Senior Capstone, primary instructor (19 students)
 2024 Summer CSCI 4950 Study Abroad - Japan, primary instructor (4 students)
 2024 Summer CSCI 4712 Senior Capstone, primary instructor (4 students)
 2024 Spring CSCI 4712 Senior Capstone (28 students)
 2023 Fall CSCI 4990 Undergraduate research, primary instructor (2 students)
 2023 Fall CSCI 4712 Senior Capstone, primary instructor (11 students)
 2023 Fall - Advising 1x full time MS and 1x part time PhD students
 2023 Research committee for external PhD student, Natalie Tracy
 2023 Summer - Co-advised 2x students (Bonn-U and UC-Irvine) during their studies at Osaka-U
 2023 Summer - CSCI 4950 Special Topics in Virtual Reality - Study Abroad (3 students)
 2023 Spring - Advising 1x MS and 1x part time PhD Students
 2023 Spring - Hired/Advised 9x total Undergraduate Research Assistants
 2023 Spring - CSCI 4712 Senior Capstone, primary instructor (27 students)
 2022 Fall - CSCI 4712 Senior Capstone, primary instructor (12 students)
 2022 Fall - CSCI 6910 Master's Thesis Research, primary instructor (1 student)
 2022 Fall - CSCI 8720 Problems in Computer & Cyber, primary instructor (1 student)
 2022 Fall - CSCI 7950 Selected Topics in Virtual Reality (1 student)
 2022 Fall - Advising 1x MS and 1x part time PhD students
 2022 Fall - Hired/Advised 6x total Undergraduate Research Assistants
 2022 Fall - Teaching peer review by Ahmed Aleroud
 2022 Spring - Advising 1x MS student at AU
 2022 Spring - Hired/Advised 2x Undergraduate Research Assistants
 2022 Spring - CSCI 4711 Software Engineering - Primary Instructor (12 students)
 2022 Spring - CSCI 4712 Capstone group co-advisor (4 students)
 2021 Hired/Advised 1x Undergraduate Research Assistant
 2021 Fall - AIST 2220 Intro. Web Development - Primary Instructor (30 students)
 2021 Fall - CSCI 4712 Capstone - Co-advisor (two teams)
 2021-2022 Continued advising of grad students (2x PhD, 1x MS, 2x co-advised groups) at Osaka-U

Service to Augusta University / SCCS / community:

2025-2026 Provost Learning Community, Section Leader: "Safe Automation with AI"

2025 Invited Speaker on AI at Via Cognitive Health
2025 ISMAR Doctoral Consortium Mentor
2024 AU Cyber Sculpture Committee
2024 SCCS Research Infrastructure Committee
2021-2025 Girls Who Code Faculty Advisor
2023-2025 Game Design Club Faculty Advisor
2023-2025 AU International Study Abroad Committee, SCCS representative
2024 Invited speaker at Brandon Wilde
2023 Invited speaker at the Augusta Kiwanis Club
2023 Wheeler High School Magnet Program summer research project mentor (Kavin Prabhakar)
2023 SCCS Research Infrastructure Committee
2022-2023 Girls Who Code Faculty Advisor
2022-2023 Game Design Club Faculty Advisor
2023 Greenbriar High School senior project mentor (Thaniel Raynack)
2022 Guest Lecturer and Journal Club moderator for Neuroscience PhD (NURO 8082)
2022 Interviewer for Nursing Department Faculty Candidates (Hope Bussenius, Kyrá Mendez)
2022 Interviewer for Neuroscience Department Faculty Candidates (Michael Goard)
2022 Augusta Brain Aging & Neurodegeneration Symposium Invited Speaker
2022 August Riverwalk Cleanup Organizer
2022 SCCS Game Design Club Faculty Advisor
2021-2022 SCCS Graduate Student Recruiting Committee
2021 Guest speaker (remote) at The University of Auckland Empathic Computing

Honors and Awards

Best Poster Award - IEEE Virtual Reality 2024
Honorable Mention for Best Paper Presentation - IEEE Virtual Reality 2024
Outstanding Faculty Award - Augusta University - June 2023
Osaka University Award - Osaka University - November 2020
Boundless Teaching Award - Augusta University - November 2020
Kasami Award - Osaka University Graduate School of Science and Technology - March 2018
ACM TiiS Board of Distinguished Reviewers - 2018
Best Paper: Augmented Human - February 2016
Best Poster: The 9th Young Researcher's Retreat, Nara, Japan. 2015
Best Paper: International Conference on Intelligent User Interfaces - March 2013
Best Research Presentation: Osaka University - August 2012
Technology Engineer of the Year: McKesson Corporation - April 2009
President's Undergraduate Research Awards: Georgia Institute of Technology - spring and fall 2006

Invited Talks

Tohoku University 2026 – Invited Lecturer – Information Content Course
OA Parks Symposium 2025 - Invited Panelist
SCCS Fall Fest 2024 – Invited Panelist
OA Parks Symposium 2023 - Invited Speaker
Guest speaker at the 2022 AU Department of Neuroscience and Regenerative Medicine
IEEEVR 2017 - PERCAR Workshop - Invited Speaker
ISUVR 2017 - Invited Speaker

Committees and Chairing (Service)

2026 - ISMAR Doctoral Consortium Chair
2025 - AIXVR 2026 Program Chair
2025 - NSF Panel: Human Centered Computing Reviewer
2025 - IEEEVR Student Volunteer Co-Chair
2024 - IEEE ISMAR Program Committee
2024 - IEEEVR Student Volunteer Co-Chair
2023 - ISMAR Journal Track Review Continuity Committee

2023 - VRST Publications Co-chair
 2023 - ISMAR Journal Track International Program Committee
 2023 - IEEEVR Doctoral Consortium Mentor
 2023 - IUI Program Committee
 2023 - Shonan Seminar #174 Co-Organizer
 2022 - AIVR Special Sessions and Late-breaking-results Co-program Chair
 2022 - ISMAR International Program Committee
 2022 - IEEEVR Online Experience Chair
 2022 - Computers and Graphics, Managing Guest Editor
 2021 - ICAT-EGVE Paper (Program) Chair
 2021 - ISMAR Doctoral Consortium Chair
 2021 - IEEEVR Online Experience Chair
 2020 - AIVR Program Chair
 2020 - ISMAR International Program Committee
 2020 - IEEEVR Dissertation Award Committee
 2020 - IEEEVR VR in VR Chair
 2019 - IEEEAIVR Program Committee
 2019 - IEEEVR Doctoral Consortium Chair
 2018 - ISMAR Doctoral Consortium Chair
 2018 - ISMAR International Program Committee
 2018 - IUI Doctoral Consortium Chair
 2018 - IUI Poster Program Committee
 2018 - IUI Program Committee
 2017 - IUI Program Committee
 2016 - IUI Program Committee
 2013 - VRSJ Publication Chair Assistant (Website)

Conference/Journal/Grant Reviewing

National Science Foundation (HCC Directorate)
 Computer Supported Cooperative Work (CSCW)
 Intelligent User Interfaces (IUI)
 International Symposium on Mixed and Augmented Reality (ISMAR)
 International Conference on Artificial Teleexistence (ICAT)
 User Interface Software and Technology (UIST)
 Virtual Reality Software and Technology (VRST)
 IEEE Virtual Reality (IEEEVR)
 Augmented Human (AH)
 Spatial User Interaction (SUI)
 Computer Human Interaction (CHI)
 Frontiers in Virtual Reality
 Army Research Laboratory (ARL)
 Artificial Intelligence and Virtual Reality (AIVR)
 ACM Transactions on Intelligent Interactive Systems (TiiS)
 Transactions on Visualization and Computer Graphics (TVCG)
 Tangible, Embedded, and Embodied Interaction (TEI)
 Presence: Teleoperators and Virtual Environments
 Interactive, Mobile, Wearable and Ubiquitous Technologies (IMWUT)
 MobileHCI

Graduate Students Supervised or Currently under Supervision

Jeffrey Benson (PhD, current)
 Alireza Taheritajar (PhD, current)
 Mo Zihao (PhD) (committee, current)

Lynsey Steinberg (PhD, current)
Katrina Visintainer (PhD, current)
Rwan Hayek (MD) (co-supervised)
Furkan Kalisir (MS) (co-supervised)
Guanghan Zhao (PhD)
Sakamoto Kenya (PhD) (co-supervised)
Adam Mansour (MS)
Nermin Shaltout (PhD)
Brandon Huynh (PhD)
Qifan Wu (MS)
Peter Kim (MS)
Lance Zhang (MS)
Patrik Goncalves (MS) (co-supervised)
Ryosuke Goto (MS)
Chang Liu (PhD)
Sophia Onyshko (MS)

Undergraduate Research Assistants Supervised or Currently Under Supervision

Anthony Gibson (BS, current)
Rhett Elliot (BS, current)
Beyza Koseoglu (BS)
Carly Whitehead (BS)
Xavier Mitchell (BS)
Martin Navarrete (BS)
Logan McCormick (BS)
Jeff Benson (BS)
Cameron Boyd (BS)
Genevieve Lucas (BS)
Erkin Khoroshi Eisalou (BS)
Kaylee Andrews (BS)
Mia Thomas (BS)
Layla Erb (BS)
Brandon Wilburn (BS)
Maddie Smith (BS)
Jamie (James) Navarro (BS)
Katrina Visintainer (BS)